

CompositionProof-2 — Preregistration

Experiment: CompositionProof-2 — multi-epoch cross-witness correlation

Program: Remnant Fieldworks Inc. — ExecutionProof™ / CIF, “Proof composition” candidate law (Deep Research Question #5)

Author: Hone, Derek — Remnant Fieldworks Inc.

Status: PREREGISTERED — design frozen BEFORE any hardware result. ZERO QPU spent at time of registration.

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This document is written and SHA-256 locked **before** any hardware epoch is collected. The classification rule (§4), power analysis (§5), permitted claim (§6) and forbidden claims (§7) are fixed in advance so the eventual verdict cannot be steered by the data. Hardware is gated behind an explicit CEO authorization token `GO COMPOSITIONPROOF-2` (see §9).

1. Question and motivation

CompositionProof-1 and OMNI-1 (DOI 10.5281/zenodo.21436092) each certified that multiple nonclassicality witnesses pass **simultaneously in one recorded state** (a single calibration epoch). Neither could address a distinct, genuinely new question, because both are **single-epoch**:

When the same two witnesses (CHSH and Mermin-3) are executed repeatedly across many DISTINCT calibration epochs on the same device, are their epoch-to-epoch FLUCTUATIONS statistically correlated?

This is the empirical content of the RF “Proof composition” candidate law:

- **Correlated fluctuations** → the two witnesses share a **common-mode device dependence**; composition on one chip is not a clean combination of independent evidence channels.
- **Uncorrelated fluctuations** → the witnesses drift independently; composition behaves like independent evidence channels **within tested bounds**.

CompositionProof-2 does not redesign the witnesses. It is the **multi-epoch extension of CompositionProof-1’s exact witnesses**.

2. Witnesses (reused verbatim from CompositionProof-1)

Per epoch, one 8-circuit battery, **byte-compatible** with CompositionProof-1:

Witness	Circuits	Statistic	Single-epoch pass threshold
W1 CHSH (2-qubit Bell)	4 (a0_b0, a0_b1, a1_b0, a1_b1)	S	$S \geq 2.2$
W2 Mermin-3 (3-qubit GHZ)	4 (XYY, YXY, YYX, XXX)	M	$M \geq 2.6$

- Shots per circuit: **4000** (frozen).
- The circuit definitions are identical to `submit_epoch.py::build_all_circuits` and `composition2_sim.py`; any change breaks the SHA-256 manifest.

3. Longitudinal design

- **One 8-circuit battery per DISTINCT device calibration epoch.**
- **“Epoch” is defined as a distinct device calibration state**, identified by `backend.properties().last_update_date`. Two batteries submitted inside the same calibration cycle are **NOT** two independent epochs and are rejected by the submitter and the verifier.
- **N_target = 30** distinct calibration epochs. **N_min = 20**; below N_min the verdict is capped at INCONCLUSIVE.
- Because IBM devices recalibrate roughly once per day, ~30 epochs span **weeks** and cross **multiple 28-day rolling QPU windows**. The ~330 QPU-seconds of total sampler time is therefore spread across windows and does **not** consume 330 s from any single window.
- Each epoch records a calibration snapshot (calibration timestamp, median T1, median T2, median readout error) so independence is auditable after the fact.

4. Estimator and FROZEN classification rule

For each epoch i the battery yields (S_i, M_i) . Over the N epochs:

- **Primary estimator:** Spearman rank correlation ρ_s between $\{S_i\}$ and $\{M_i\}$.
- **Secondary estimator:** Pearson correlation r (reported, not decisive).
- **Uncertainty:** 95% **epoch-bootstrap** CI on ρ_s (resample epochs with replacement, `N_BOOT = 5000`, `BOOT_SEED = 20260908`).
- **Material-effect band:** `EFFECT_BAND = 0.30` ($|\rho| \geq 0.30$ is material).
- **Shot-noise drift guard:** the epoch-to-epoch std of each witness must exceed the shot-noise floor $2/\sqrt{\text{shots}}$ by `DRIFT_GUARD_Z = 2.0`. You cannot estimate the correlation of fluctuations that are indistinguishable from shot noise.

Frozen verdicts (evaluated in this order):

Verdict	Condition
INCONCLUSIVE_NO_DRIFT	either witness's epoch std does not clear the drift guard (no real drift to correlate).
INCONCLUSIVE	$n_{\text{eff}} < N_{\text{min}} (20)$, OR epochs not independent (shared calibration timestamps), OR the bootstrap CI straddles both 0 and $\pm \text{EFFECT_BAND}$.
CORRELATED	drift present, epochs independent, $n_{\text{eff}} \geq N_{\text{min}}$, and the 95% CI for ρ_s lies entirely outside ($-\text{EFFECT_BAND}$, $+\text{EFFECT_BAND}$) (i.e. $ \rho_s $ materially and significantly > 0).
UNCORRELATED	drift present, epochs independent, $n_{\text{eff}} \geq N_{\text{min}}$, and the 95% CI for ρ_s lies entirely inside ($-\text{EFFECT_BAND}$, $+\text{EFFECT_BAND}$) (equivalence: correlation certified small).

These constants are duplicated in `verifier2.py` and `composition2_sim.py` and are covered by the SHA-256 manifest.

5. Power analysis and two-stage adaptive plan

Control simulations (`control_validate2.py` , output `control_validation2_output.json`) establish an important, honest asymmetry:

- **Detecting** a real correlation (e.g. true $\rho \approx 0.75$) is reliable at **$n \approx 30$** .
- **Certifying ABSENCE** of correlation — the UNCORRELATED equivalence verdict, CI entirely within ± 0.30 at true $\rho = 0$ — requires roughly **$n \approx 150$** epochs.
At $n = 30$ with true $\rho = 0$ the honest verdict is INCONCLUSIVE, not UNCORRELATED.

Preregistered **two-stage adaptive design**:

- **Stage 1 — $n = 30$ epochs.** If the verdict is **CORRELATED**, stop: the primary hypothesis (common-mode composition) is supported.
- **Stage 2 — extend toward $n \approx 150$ epochs** ONLY if Stage 1 is INCONCLUSIVE and we wish to certify UNCORRELATED. Stage 2 is itself gated (a fresh `GO COMPOSITIONPROOF-2` is required to continue collecting epochs).

6. Permitted claim (the ONLY claim this experiment can support)

Across N distinct measured calibration epochs on one named IBM device over the measured window, the epoch-to-epoch fluctuations of the CHSH and Mermin-3 witnesses were [correlated / uncorrelated within ± 0.30 / inconclusive], with the stated Spearman estimate and 95% epoch-bootstrap CI.

7. Forbidden claims (explicitly out of scope)

- **X** “Proved the witnesses are independent” (absence needs the Stage-2 equivalence result, and even then only within ± 0.30 on one device).
- **X** “Proved a causal common cause” (correlation $\neq$ causation).
- **X** “Device-independent” or “loophole-free” (locality + detection loopholes OPEN).
- **X** “Universal” / “for all devices” / “for all witness pairs” (one device, one pair, one window).
- **X** “First / only” anything beyond what is documented by cited DOIs.

8. Honesty bounds (carry into every artifact)

- Device-dependent witnesses on **co-located qubits**. Locality and detection loopholes **OPEN**. NOT device-independent; NOT loophole-free.
- Any cross-epoch correlation is an **OBSERVED association** on one device over the measured window — not a causal common cause, not a universal statement.
- **“Epoch” = a distinct calibration state**. Batteries sharing a calibration cycle are not independent epochs.
- The Aer simulation is a **feasibility check only** (that the estimator and guard behave), not evidence about real hardware correlation.
- Correlation $\neq$ causation; a small measured correlation does not prove mechanistic independence.

9. QPU budget, gating, and timeline

- **Per epoch:** 8 circuits $\times$ 4000 shots $\approx$ ~ 11 QPU-seconds of sampler time (backend-dependent).
- **N_target = 30 epochs $\rightarrow$ ~ 330 QPU-seconds total**, spread across weeks and multiple 28-day rolling windows (≈ 1 epoch per calibration cycle).
- **Hardware is GATED.** `submit_epoch.py` submits nothing unless ALL gates pass: `--go-token "GO COMPOSITIONPROOF-2"`, `--i-have-ceo-authorization`, a valid `--epoch-id`, and a verifying SHA-256 manifest. It additionally refuses to record an epoch whose calibration timestamp already appears in the series (independence guard).
- At preregistration time: **hardware_submitted = false; 0 QPU-seconds spent.**

10. Reproducibility artifacts

File	Role
<code>composition2_sim.py</code>	Aer feasibility simulation (ZERO QPU).
<code>verifier2.py</code>	INDEPENDENT verifier (SciPy; distinct implementation).
<code>control_validate2.py</code>	Synthetic control suite (all verdicts reachable).
<code>submit_epoch.py</code>	GATED per-epoch hardware submitter.
<code>composition2_sim_result.json</code>	Feasibility sim output.
<code>control_validation2_output.json</code>	Control suite output (all_pass).
<code>series_example_sim.json</code>	Example multi-epoch series (from sim).
<code>verifier2_output_example.json</code>	Independent verifier run on the example.
<code>SHA256_MANIFEST.txt</code>	Freezes every artifact above + this document.
<code>lock_candidate.json</code>	Composite lock + design summary.